



SIMATS ENGINEERING

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Course Code: CSA1017

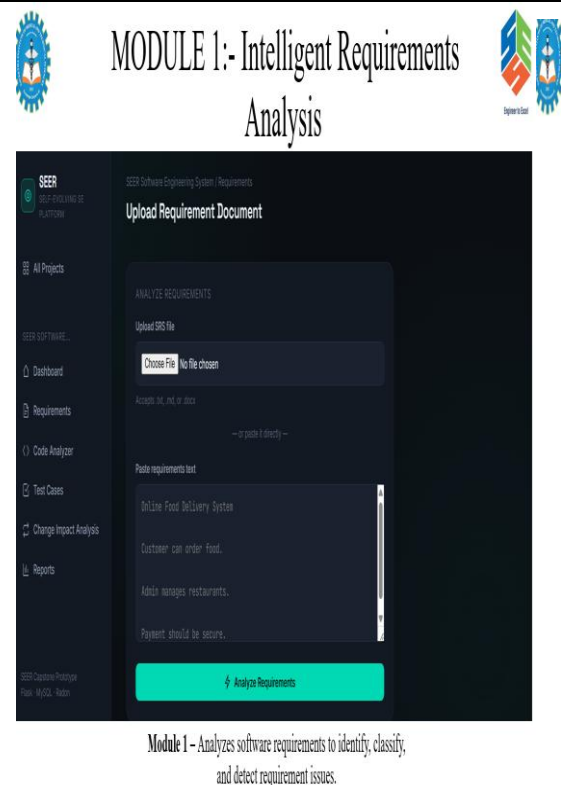
Slot: C

Course Name: SOFTWARE ENGINEERING

Course Faculty: Dr. LATHA R

Project Title: Self-Evolving Engineering Ecosystem for Resilient Software Systems

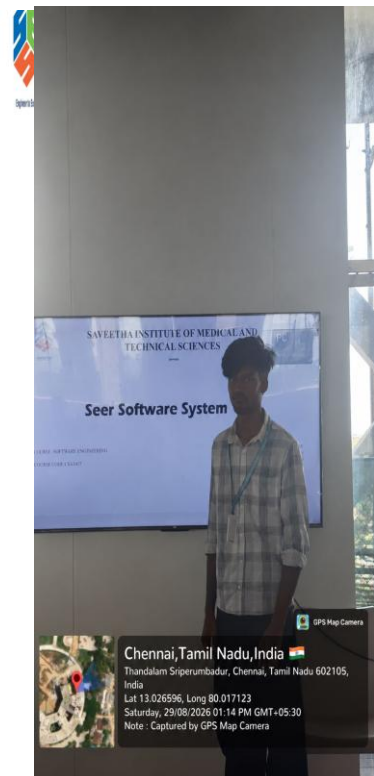
Module Photographs:



MODULE 1:- Intelligent Requirements Analysis

Requirements Analysis

- Upload .txt, .md or .docx requirements.
- Extract requirements from the document.
- Identify requirement information.
- Generate requirement reports.
- Generate test cases from requirements.



Project Description:

MODULE 1: PREPROCESSING AND FREQUENCY ANALYSIS

Module 1 focuses on preparing the AI dataset for efficient compression using Huffman Coding. The module accepts the dataset as input and performs preprocessing to organize the data before compression. It identifies the unique characters or symbols present in the dataset and calculates the frequency of each symbol. A frequency table is generated to show how often each character occurs. The module also calculates the Shannon entropy of the dataset to understand its information content and estimate the efficiency of compression. The analysed frequency information is then provided to the next module for constructing the Huffman tree and generating binary codes. This module is important because Huffman Coding assigns shorter codes to frequently occurring symbols and longer codes to less frequently occurring symbols. Thus, Module 1 provides the required input and analysis for achieving efficient and lossless dataset compression.

Student Signature

Guide Signature